

(at 300 K, $J \approx 20$ are needed to converge the rate constant). However, these results do show resonances measured in milliwavenumbers are primarily responsible for deviations from experiment. If such resonance widths were to be lightly sampled in the full spectrum of resonances for all important values of J , then experiment and theory could converge on a negligible pressure dependence in the observable range. The calculated resonance widths and locations are numerically exact given the potential energy surface. The location and widths of resonances can be a very sensitive function of the potential energy surface. The calculated resonance locations are known⁹ to systematically deviate from spectroscopic measurements. Thus, a corrected surface could produce results that even for $J = 0$ could be consistent with experiment. This is a return to the notion advanced by Pritchard¹⁹ specifically in regards to HCO that potential energy surface features can especially influence the low-pressure limit in the case of isolated resonances.

No amount of reasonable tampering with the potential energy surface will remove the barrier to addition, a feature consistent

with the available kinetics experiments.^{2,15,16} This means that there will be long-lived resonances at low energies that receive large Boltzmann weights. Such resonances will be prominent for low values of J but less so for high values of J , where the centrifugal potential has largely filled in the well region of the potential energy surface. Resonances for high J will come in $2J + 1$ member families (different K quantum numbers) but will also be higher in total energy and receive a lower Boltzmann weight. While nothing quantitative can be said, it would appear likely that very low pressure studies of H + CO addition would reveal resonance behavior through pronounced changes in the apparent low-pressure limit. Such features might very well change with temperature, becoming apparent up to a higher pressure as the temperature decreases. In this sense, both the low-pressure limit in its approach and the high-pressure limit in its value reveal through kinetics experiments important characteristics of resonances.

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On the Question of Negative Activation Energies: Absolute Rate Constants by RRKM and G1 Theory for $\text{CH}_3 + \text{HX} \rightarrow \text{CH}_4 + \text{X}$ ($\text{X} = \text{Cl}, \text{Br}$) Reactions

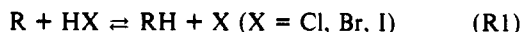
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The kinetics and mechanisms of the reactions $\text{CH}_3 + \text{HX} \rightarrow \text{CH}_4 + \text{X}$ ($\text{X} = \text{Cl}, \text{Br}$) were investigated. Both reactions proceed via a loosely bonded complex which is formed without activation energy. Rate constants of each reaction have been calculated based on RRKM theory with corrections for tunneling evaluated using the Wigner method. For the chloro system, a quantitative agreement is found between theoretical results and available experimental data. In the case of the bromo system, the theoretical barrier must be lowered to -0.11 kcal/mol to match experimental results. Isotope effects are predicted; $k_{\text{H}}/k_{\text{D}}$ values follow (296 K): $\text{CH}_3 + \text{DX}$, 5.08 ($\text{X} = \text{Cl}$), 1.98 ($\text{X} = \text{Br}$); $\text{CD}_3 + \text{HX}$, 0.62 ($\text{X} = \text{Cl}$), 0.87 ($\text{X} = \text{Br}$); $\text{CD}_3 + \text{DX}$, 3.08 ($\text{X} = \text{Cl}$), 1.62 ($\text{X} = \text{Br}$). Quantitative agreement is found for the single experimental value at 296 K for $\text{CD}_3 + \text{HBr}$ vs $\text{CH}_3 + \text{HBr}$: $k_{\text{H}}/k_{\text{D}} = 0.83 \pm 0.08$.

Introduction

Kinetic and equilibrium studies of halogenation reactions of the type



have been a major source of information on polyatomic free-radical heats of formation ($\Delta H_f^\circ(\text{R})$). The majority of such thermochemical investigations have employed bromination and iodination reactions.^{1,2} In principle, such studies involve the determination of the enthalpy change, $\Delta H_{\text{R1}}^\circ(\text{T})$, for the reaction R1 and combine this information with the known heats of formation of other species. In the second-law method, $\Delta H_{\text{R1}}^\circ(\text{T})$ is obtained directly from the difference in activation energies ($E_f - E_r$) for the forward and reverse reactions, preferably at the midpoint of the overlapping temperature range. In the alternative third-law method, $\Delta H_{\text{R1}}^\circ(\text{T})$ is obtained from knowledge of the forward and reverse rate constants (or their ratio, k_f/k_r) and the entropy change of reaction at some particular temperature; thus $\Delta H_{\text{R1}}^\circ(\text{T}) = \Delta G_{\text{R1}}^\circ(\text{T}) - T\Delta S_{\text{R1}}^\circ(\text{T})$, where $\Delta G_{\text{R1}}^\circ(\text{T}) = -RT \ln K_p(\text{T}) = -RT \ln (k_f/k_r)$. In either case, heat capacity data (to correct to 298 K), and in the third-law method, the entropies of the species, must be known or derived from structural data.

Reactions of polyatomic radicals with hydrogen halides are known to be very rapid¹ with "near zero" activation energies. Thus, earlier classical studies (including numerous competitive studies) have mainly focused on the determination of the rate constants for the reverse reactions, namely H abstraction by halogen atoms.³ For the evaluation of thermochemical data, a small positive activation energy characteristic of HX was then assigned for the forward reaction, R1. The assumed activation energies (for any R) have been 1 ± 1 kcal/mol for $\text{R} + \text{HI}$ and about 2 ± 1 kcal/mol for $\text{R} + \text{HBr}$.^{1,2} However, in recent years, a number of investigators⁴⁻⁸ have called attention to the fact that heats of formation of alkyl radicals derived in this manner are, in general, lower than the $\Delta H_f^\circ(\text{R})$ values derived from other equilibria not

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involving the halogens, such as radical dissociation and recombination reactions:



A comprehensive analysis of such systems for $R = C_2H_5$, $n-C_3H_7$, $i-C_3H_7$, $sec-C_4H_9$, and $t-C_4H_9$ has been reported by Tsang.^{4,5} The most noteworthy feature of this upward revision of $\Delta H_f^\circ(R)$ is that at the upper end of the range of values they predict negative activation energies for reaction R1 for $X = Br$ or I (but not for $X = Cl$).

Most recently, in a series of papers, Gutman and co-workers reported directly measured rate constants with negative temperature coefficients for the alkyl radicals: CH_3 , C_2H_5 , $i-C_3H_7$, $sec-C_4H_9$, and $t-C_4H_9$ with HBr ⁹⁻¹² and with HI .¹³ The kinetics of the reactions were studied in a tubular reactor coupled to a photoionization mass spectrometer. Although negative temperature dependences are not uncommon for complex reactions, the results of Gutman and co-workers were, nevertheless, unexpected, in that they apply to processes previously believed to be pure metathesis reactions. Therefore, it is not surprising that they have been received with some skepticism.^{14,15} However, the negative temperature dependence for the reaction of $t-C_4H_9$ with HBr , DBr , and HI appears to have been confirmed in independent time-resolved experiments by following the production of isobutane using a tunable diode infrared laser probe.¹⁶ In a followup study, Seetula and Gutman¹⁷ also measured negative activation in reactions of some halogen-substituted methyl radicals with HI , with the interesting observation of a transition from small positive activation energies for $CH_2Cl + HI$ and $CHCl_2 + HI$ to small negative activation energies for CH_2Br and CH_2I with HI , respectively. Analogous to CH_3 , negative activation energies were also found in a study of the reaction of SiH_3 with HBr and HI , which combined excimer laser flash photolysis with photoionization mass spectrometry.¹⁸ The heatable tubular reactor/photoionization mass spectrometry technique has also been used to study the kinetics of some of the aforementioned alkyl radicals with molecular bromine.¹⁹ In all cases, small negative activation energies have been measured.¹⁹ By inference, the implication of these findings is that activation energies of all $R + I_2 \rightarrow RI + I$ reactions are also negative, instead of being zero, as has been previously assumed and extensively used in the determination of bond dissociation energies.^{1,2}

Gutman and co-workers^{9,10,13} interpreted the negative activation energies for reaction R1 in terms of a complex reaction mechanism in which the alkyl radical is initially attracted to the halogen end of HX to form a quasi-stable complex, followed by rotation of the hydrogen. McEwen and Golden²⁰ have carried out an approximate RRKM calculation for $t-C_4H_9 + HI \rightarrow i-C_4H_9 + I$ that models the reaction as a two-channel chemical activation system involving a $t-C_4H_9 \cdots HI$ collision complex, following the theoretical model set forth by Mozurkewich and Benson.²¹ Their results show that based on their assumed model, a negative tem-

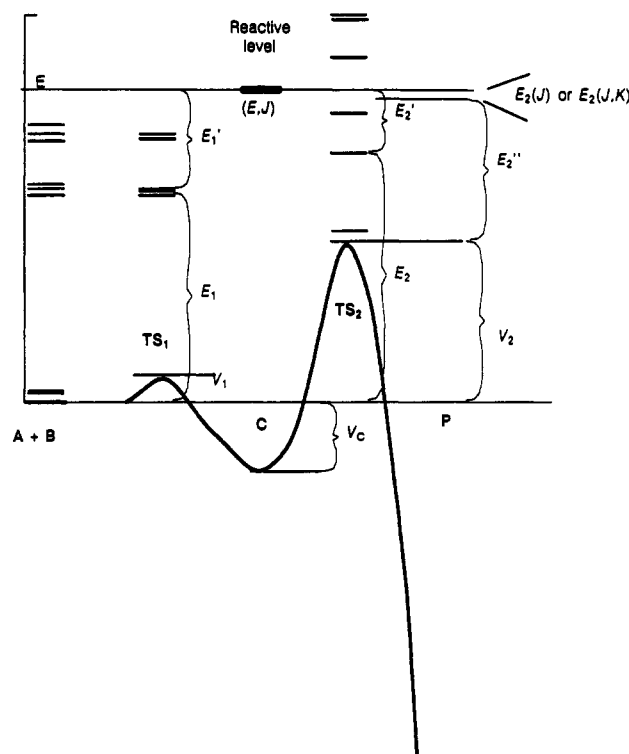
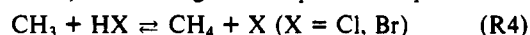


Figure 1. Generic potential energy surface for a bimolecular reaction in which an intermediate complex is formed.

perature dependence can be rationalized, but the theoretical treatment cannot resolve the fundamental question as to whether this reaction is a metathesis or chemical activation system. However, based on the model selected, the calculations of McEwen and Golden predict an inverse primary kinetic isotope effect ($k_H/k_D < 1$, decreasing with temperature for $t-C_4H_9 + HI/DI$) which they suggest as a criterion to discern the two cases.

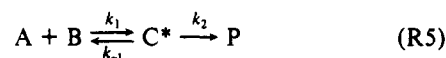
In the present work, we apply a variation of Mozurkewich and Benson's RRKM theory to the reactions R4 ($X = Cl, Br$), one of which ($X = Br$) exhibits negative temperature dependence.¹⁰



The structures of all species, including the intermediate complex and the transition state for its decomposition to products, have been determined theoretically, as have vibrational frequencies.²² The potential surface for the reaction has been determined at the highest available level of theory.²² The results of direct computation of the rate constants and kinetic isotope effects are presented below. We begin, however, with a detailed description of the procedure for arriving at the theoretical rate constants.

Intermediate Complex and Negative Activation Energies

The following is a derivation of a general expression for a reaction rate constant for a bimolecular reaction in which an intermediate complex is formed. The analysis parallels closely that of Mozurkewich and Benson (MB)²¹ and is an extension of RRKM theory.²³ The reaction is analyzed according to the general equation



in which two reactants A and B are assumed to have enough energy to form products P on a reaction potential surface such as shown in Figure 1. In the present case, A is a CH_3 or CD_3 free radical and B is HX or DX . An intermediate complex C corresponding to a shallow well in the potential surface is formed. Such a complex is indeed found for both halo systems considered

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here. In the low-pressure limit, C is formed "hot", hence C^* . A transition structure TS_2 hinders the decomposition of C to products with a potential barrier V_2 . The formation of C is hindered by a potential barrier V_1 . While in the present cases $V_1 = 0$, following MB, we postulate, nevertheless, a "transition state" TS_1 for the formation process. The geometry of TS_1 is not well-defined in the sense that it corresponds to a stationary point (with one imaginary frequency) on the potential surface. However, it corresponds to a geometry that "is on the way to forming C". In other words, thinking in terms of the reverse reaction, the "transition state" would correspond to an extension of the reaction coordinate to a point where only a weak interaction between the incipient fragments A and B remained, perhaps balanced by centrifugal forces. For the purpose of wider generality, we present the formulation with a barrier V_1 , which may be zero or positive.

The vibrational and rotational states of TS_1 constitute a filter for the transformation of $A + B$ to C and vice versa. These are not the same as the vibrational and rotational states of the separate species, A and B, although close similarities are to be expected. Likewise, the vibrational and rotational states of TS_2 constitute a filter for the further reaction of C to products P. Angular momentum and total energy are assumed to be conserved during the course of the reaction. The reaction will be analyzed in a center of mass coordinate system; the 3 translational degrees of freedom will be factored out of all partition functions, and the kinetic energy associated with translation of the center of mass is kept constant. The total energy of species A and B shown as E in Figure 1 is distributed as kinetic energy of relative motion of A and B (the external part), as well as vibrational and rotational motions (the internal part) of both species. The internal part of the kinetic energy is quantized, the vibrational kinetic energy being characterized by separate quantum numbers $\{\nu_A\}$ and $\{\nu_B\}$ and the rotational kinetic energy being specified by a "pseudo" quantum number J^{int} , which classically would be the vector sum of the angular momenta J_A and J_B of A and B, respectively. The relative motion of A and B has a rotational component, and hence an associated angular momentum, which is also quantized, and characterized by quantum number J^{ext} . Classically, the total angular momentum of the system J , which would be conserved throughout the reaction, is the sum of these: $J = J^{ext} + J^{int}$. In the limit that A and B are far apart compared to their internal dimensions, the $A \cdots B$ system may be approximated instantaneously as a two-body rigid rotor. The energy associated with external rotation will be given by $E(J^{ext}) = B_{A+B} J^{ext}(J^{ext} + 1)$, where the rotational constant B_{A+B} is very small and hence the part of the kinetic energy associated with "external" rotation is virtually zero. The formal degeneracy of the "levels" is $2J^{ext} + 1$. The important difference between the separated $A-B$ system and TS_1 is that the *external* part of the angular momentum has become better defined and a significant fraction of the external part of the kinetic energy is now in the form of rotational kinetic energy, quantized according to $E(J^{TS_1}) = B_{TS_1} J^{TS_1}(J^{TS_1} + 1)$. It will be assumed that the *internal* rotational kinetic energy of A and B is unchanged in TS_1 . Passage from TS_1 to C is accompanied by loss of independent rotations of the fragments, the angular momentum associated with that motion being added to the *external* angular momentum to yield the angular momentum of C, which is characterized by quantum number J . Since C is a stable intermediate, all of its states are assumed to be quantized at the energy at which it is formed, with the exception of the internal degree of freedom (a vibrational mode) corresponding to the reaction coordinate. The latter is quantized only up to the lowest channel out of the potential well. The reaction to form products is analyzed from the point of view of the options available to the intermediate C.

Reaction of the Intermediate C. For the purpose of the following and subsequent discussion, we will refer to all energies as relative to the lowest energy available to the separated species A and B. Thus, the lowest energy of C is a negative quantity V_C . The energy of the first transition state V_1 is a positive quantity or zero. The energy of the second transition state V_2 (Figure 1) may be either positive or negative but greater than V_C . Let us suppose that C has been prepared in a state, $C(E, J)$, specified by its energy E

and its rotational quantum number J . Clearly, the lowest possible value of E achievable from reactants is V_1 . Within the context of RRKM theory, "active" modes are those modes which lose their identity during the course of the reaction. Redistribution of the energy among the "active" modes of C is assumed to be rapid. Thus, while $C(E, J)$ has enough energy to revert to reactants A and B, it does not remember the path of its creation and must find again the "windows" of TS_1 . Forward options available to $C(E, J)$ depend on the relative magnitudes of V_1 and V_2 and the value of E relative to the latter barrier. If the second barrier is higher ($V_2 > V_1$) and E is less than V_2 , the exit channel back to reactants is available, but $C(E, J)$ may only react to form products P by tunneling through the barrier. If E is equal to or greater than V_2 , both classical channels are available. Since the exit channels, via TS_1 and TS_2 , are at least partially quantized, only states of both whose total energy is less than or equal to E and whose total angular momentum is the same may serve as conduits for the decomposition of $C(E, J)$. Because of the rapid redistribution of internal energy, all states of TS_1 with positive energy less than or equal to E and greater than V_1 are available to $C(E, J)$ for the retro reaction, subject to the conservation of angular momentum. If a discrete state of TS_1 with energy E_1 less than E is used as the exit channel, the energy discrepancy, $E_1' = E - E_1$, is assumed to be taken up as kinetic energy of motion along the reaction coordinate. Since this is not quantized and since redistribution of the internal energy among the vibrational modes is assumed to be very rapid, there are no *energetic* impediments to the involvement of states whose remaining modes are quantized. The statistical probability of the involvement of states at any discrete energy level E_1 with angular momentum J is simply taken to be proportional to $N_1(E_1, J)$, the number of states of TS_1 with those characteristics. The total rate of decomposition of $C(E, J)$ via TS_1 is therefore proportional to the "concentration", $[C(E, J)]$, and the total number of states that exist with energy less than E but greater than V_1 and with angular momentum J . The proportionality constant will depend on both E and J . Thus, the rate constant for the reverse process is given by

$$k_{-1}(E, J) = K(E, J) W_1(E, J) \quad (1)$$

where the proportionality constant is given by RRKM theory

$$K(E, J) = 1/h N_C(E, J) \quad (2)$$

where h is Planck's constant and the quantity $N_C(E, J)$ is the number of states available to C at energy E which have angular momentum J . The remaining quantity

$$W_1(E, J) = \sum_{E_1=V_1}^E N_1(E_1, J) \quad (3)$$

and the treatment of the angular momentum J will be discussed further below. Exactly parallel statements may be made for the forward reaction of $C(E, J)$ and analogous quantities defined. Thus

$$k_2(E, J) = K(E, J) W_2(E, J) \quad (4)$$

where

$$W_2(E, J) = \sum_{E_2=V_2}^E N_2(E_2, J) \quad (5)$$

The *classical* rate of formation of products at any specified E, J is equal to $k_2(E, J)[C(E, J)]$, and the total *classical* RRKM rate of product formation is obtained by summation over all possible energies and values of J . The actual *observed* rate of product formation should differ approximately from the classical rate by a factor κ . Thus

$$\kappa \sum_{J=0}^{\infty} \int_{V_{\max}}^{\infty} dE k_2(E, J) [C(E, J)] \approx \frac{d[P]}{dt} = -\frac{d[A]}{dt} = k_{\text{obs}}[A][B] \quad (6)$$

where $V_{\max}$ is the greater of V_1 or V_2 . The last relationship follows since the rate constant k_{obs} measures the rate of disappearance of reactants or, equivalently, the rate of formation of products. Although the factor κ may be used to hide a multitude of sins,

we take its sole function to be to correct for the absence of quantum mechanical tunneling in the classical rate expressions. It is discussed further below.

If one assumes equilibration of states of the intermediate, then the equilibrium "concentration" $[C(E,J)]_{\text{eq}}$ is related to total equilibrium concentration $[C]_{\text{eq}}$ by

$$[C(E,J)]_{\text{eq}} = \frac{N_C(E,J)[C]_{\text{eq}}}{Q_C} e^{-(E-V_C)/RT} \quad (7)$$

where Q_C is the partition function of C with the center of mass motion factored out. Since the intermediate is not actually in equilibrium, its steady-state concentration may be specified by

$$[C(E,J)] = \frac{[C(E,J)]_{\text{eq}}}{1 + \beta(E,J)} \quad (8)$$

where $\beta(E,J)$ is the branching ratio

$$\beta(E,J) = \frac{k_2(E,J)}{k_{-1}(E,J)} = \frac{W_2(E,J)}{W_1(E,J)} \quad (9)$$

The second equality in eq 9 follows from the RRKM expressions for unimolecular rate constants.²¹ Substitution of eqs 7 and 8 into eq 6 yields

$$k_{\text{obs}} \approx \kappa k_{\text{RRKM}} = \frac{[C]_{\text{eq}} e^{V_C/RT}}{Q_C[A][B]} \sum_{J=0}^{\infty} \int_{V_{\text{max}}}^{\infty} dE \frac{k_2(E,J) N_C(E,J) e^{-E/RT}}{1 + \beta(E,J)} \quad (10)$$

We note that steady-state conditions permit the following macroscopic relationships

$$\frac{[C]}{[A][B]} = \frac{k_1}{k_2 + k_{-1}} = \frac{k_1/k_{-1}}{1 + k_2/k_{-1}} = \frac{K_{\text{eq}}}{1 + \beta} = \frac{1}{1 + \beta} \frac{Q_C e^{-V_C/RT}}{Q_A Q_B} = \frac{1}{1 + \beta} \frac{[C]_{\text{eq}}}{[A][B]} \quad (11)$$

where β is the macroscopic branching ratio. Using the last equality in eq 11 and substituting eqs 2 and 4 into eq 10, one obtains

$$k_{\text{RRKM}} = \frac{1}{h Q_A Q_B} \sum_{J=0}^{\infty} \int_{V_{\text{max}}}^{\infty} dE \frac{W_2(E,J) e^{-E/RT}}{1 + \beta(E,J)} \quad (12)$$

Equation 12 is equivalent to that derived by MB²¹ for their k_{expt} . Thus, the RRKM rate constant depends only on the functions $W_1(E,J)$ and $W_2(E,J)$ and the partition functions of the reactants and not at all on any characteristics of the intermediate complex C. If the branching ratio is zero, one has the transition-state theory rate constant k_{TST} as shown by MB.²¹

$$k_{\text{TST}} = \frac{1}{h Q_A Q_B} \sum_{J=0}^{\infty} \int_{V_{\text{max}}}^{\infty} dE W_2(E,J) e^{-E/RT} = \frac{RT Q_2}{h Q_A Q_B} e^{-V_2/RT} \quad (13)$$

$W_1(E,J)$ and $W_2(E,J)$. Careful consideration must be applied to the evaluation of the functions $W_1(E,J)$ and $W_2(E,J)$, eqs 3 and 5, respectively. In RRKM theory, the states included in the sums are "active" states, i.e., those states whose kinetic energy is available for redistribution into motion along the reaction coordinate. Usually, all vibrationally excited states are assumed to be "active" in this sense. The opposite involves those states whose identity is preserved in some way. These are called "adiabatic". Rotational states are identifiable since angular momentum must be conserved in the reaction, at least in the low-pressure limit. In the case of very fast reactions, one should probably regard some of the vibrational states as adiabatic as well but this refinement is beyond the scope of the present work.

Conservation of Angular Momentum. In classical mechanics, conservation of angular momentum requires that all three vector components of the angular momentum vector remain unchanged. In quantum mechanics, a similar situation obtains, although rigorously, its implementation falls into the realm of scattering theory. Conservation of angular momentum may be imposed simplistically by insisting that one or more quantum numbers

remain fixed. For the present reaction, since TS₂ is quasi-linear, i.e., with rotational constants $A_2 \gg B_2 = C_2$, the usual approximation, that the molecule be treated as a two-dimensional (linear) rotor (i.e., J and M conserved), should work well. In this case, the energy levels depend only on J according to

$$E_2(J) = J(J+1)B_2 \quad (14)$$

and have a degeneracy of $2J+1$. The remaining rotational degree of freedom, rotation about the symmetry axis, is treated as a one-dimensional rotor with rotational constant $A_2' (\approx A_2 - C_2)$. The rotational constant A_2' should be adjusted to reproduce the correct rotational partition function. The one-dimensional rotor levels are added to the "active" levels counted by $W_2(E,J)$. It is implicitly assumed that the rotational constants do not change with the level of vibrational excitation or excitation of the one-dimensional rotor, and as a result, each "active" level has a stack of rotational levels with relative energy given by eq 14 associated with it. In this case, one has the relationship

$$W_2(E,J) = (2J+1)W_2(E_2'',0) \quad (15)$$

where $E_2'' = E - E_2(J)$ and $W_2(E_2'',0)$ is the sum of active states with $J=0$ and energy less than or equal to E_2'' (Figure 1). This is the nature of the approximation invoked by MB. By definition, $W_2(E_2'',J) = 0$ if $E_2'' < V_2$. In fact, for the forward direction, conservation of angular momentum is readily enforced because of the close similarity and nature of the structures of the complex C and the second transition state TS₂. Both are prolate symmetric rotors, with rotational constants $A_C \approx A_2 \gg B_C \approx B_2$. The energy depends on two quantum numbers, J and K , according to

$$E_2(J,K) = J(J+1)B_2 + (A_2 - C_2)K^2 \quad (16)$$

where K and the third quantum number M are independently limited by J , i.e., $-J \leq K, M \leq J$. In this case, for TS₂ we have explicitly

$$W_2(E,J,K) = D_2(J,K) W_2(E_2'',0,0) \quad (17)$$

The degeneracy $D_2(J,K)$ of each level is $2J+1$ if $K=0$ and $2(2J+1)$ if $K \neq 0$. E_2'' is given by $E_2'' = E - E_2(J,K)$, and $W_2(E_2'',0,0)$ is the sum of active states with $J=0, K=0$, and energy between V_2 and E_2'' . Clearly, $W_2(E_2'',0,0)$ counts only vibrational levels.

In the reverse reaction, dissociation of C to reactants A and B, the situation with respect to conservation of angular momentum is considerably more complicated. The total angular momentum J of the rotating complex is partitioned between rotation of the fragmenting complex as a whole J^{ext} and rotation of the individual fragments J^A and J^B . The "first transition state" TS₁ provides the window for the unimolecular fragmentation. Its active modes are taken to be the vibrational modes of the individual species A and B (7 degrees of freedom for CH₃ and HX) and a one-dimensional rotation about the symmetry axis of the assumed structure. Of the total of $3N-6$ (=15 in the present case) degrees of freedom, 1 degree of freedom corresponds to translation along the reaction coordinate. The remaining 6 degrees of freedom are taken as rotational modes of two-dimensional rotors, namely the structure as a whole, approximated as A and B separated by a distance approximately equal to or greater than the sum of van der Waals radii, with rotational constant B^{ext} and the rotations of A and B separately treated as two-dimensional rotors with rotational constants B^A and B^B , respectively. The latter modes originate from degenerate bending modes of the complex C. All values of J^{ext}, J^A , and J^B , consistent with $J = J^{\text{ext}} + J^A + J^B$ in a vector sense, are allowed provided the total energy

$$E_1(J^{\text{ext}}, J^A, J^B) = J^{\text{ext}}(J^{\text{ext}}+1)B^{\text{ext}} + J^A(J^A+1)B^A + J^B(J^B+1)B^B \quad (18)$$

is below E , the energy of C(E,J). In this treatment

$$W_1(E,J) = D_1(J^{\text{ext}}, J^A, J^B) W_1(E_1'',0) \quad (19)$$

where the degeneracy is

$$D_1(J^{\text{ext}}, J^A, J^B) = (2J^{\text{ext}}+1)(2J^A+1)(2J^B+1)S(J^{\text{ext}}, J^A, J^B) \quad (20)$$

and $E_1'' = E - E_1(J^{\text{ext}}, J^A, J^B)$. The statistical factor $S(J^{\text{ext}}, J^A, J^B)$ takes account of the fact that the angular momentum is counted as an absolute quantity and half of the rotations will be in the wrong direction. Specifically, $S(J^{\text{ext}}, J^A, J^B) = 1$ if at least two of $J^{\text{ext}}, J^A, J^B = 0$, $S(J^{\text{ext}}, J^A, J^B) = 1/2$ if any one of $J^{\text{ext}}, J^A, J^B = 0$, and $S(J^{\text{ext}}, J^A, J^B) = 1/4$ if none of J^{ext}, J^A, J^B are zero. The function $W_1(E_1'', 0)$ is equal to zero if $E_1'' < V_1$.

The absolute RRKM rate constant in the absence of tunneling, is given by

$$k_{\text{RRKM}} = \frac{1}{hQ_A Q_B} \sum_{J=0}^{\infty} \sum_{K=0}^J \int_{V_{\text{max}}}^{\infty} dE \frac{W_2(E, J, K) e^{-E/RT}}{1 + \beta(E, J, K)} \quad (21)$$

where explicit summation over both J and K is carried out and the branching ratio β is the ratio of W_2 over W_1 , specified by eqs 17 and 19. With all $\beta(E, J, K) = 0$, the transition-state theory rate constant is analogously obtained.

$$k_{\text{TST}} = \frac{1}{hQ_A Q_B} \sum_{J=0}^{\infty} \sum_{K=0}^J \int_{V_{\text{max}}}^{\infty} dE W_2(E, J, K) e^{-E/RT} = \frac{RTQ_2}{hQ_A Q_B} e^{-V_2/RT} \quad (22)$$

In practise, if the energy difference between forward and reverse barriers is greater than about 1 kcal/mol, the rotational levels of the whole disintegrating complex and of the individual fragments (i.e., TS_1) form a dense manifold of available states for the reverse reaction whereas available states for the forward reaction, at least at energies immediately above V_2 , are sparse (i.e., $W_1(E, J) \gg W_2(E, J, K)$). As a consequence, the branching ratio $\beta(E, J, K)$ will be quite small for all values of E, J , and K , and the RRKM rate constant k_{RRKM} (eq 21), will be close to the transition-state theory rate constant (eq 22). An additional consequence is that the calculated rate will be very insensitive to the choice of the structure of TS_1 . For values of $V_2 - V_1$ less than 1 kcal/mol, the number of available states for the forward reaction begins to be comparable to the manifold of states available for the reverse process, and for values of $V_2 - V_1$ close to zero or negative, especially in the low-temperature regime, substantial deviations from the transition-state theory value are expected and found. In this regime, the results are also expected to be more sensitive to the particular choice of TS_1 . More sophisticated treatments of the unimolecular decomposition process are the subject of current research²⁴ and beyond the scope of the present work. For the purpose of the present analysis, we have rather arbitrarily adopted a structure for TS_1 which places the methyl and HX moieties at a separation of 3.5 Å, i.e., just outside of van der Waals contact. This procedure yields satisfactory results for all the reactions considered.

In the present work, we evaluate k_{RRKM} (eq 21) by direct numerical integration and k_{TST} using the closed expression in eq 22. Each is multiplied by the factor κ , which we take to be a temperature-dependent tunneling correction. A number of different kinds of tunneling corrections have been evaluated by Garrett and Truhlar,²⁵ who found that the simplest, due to Wigner,²⁶ was also the most accurate. We adopt the Wigner correction

$$\kappa \approx 1 + \frac{1}{24} \left(\frac{h\nu^*}{k_B T} \right)^2 \quad (23)$$

where ν^* is the magnitude of the imaginary frequency of TS_2 and k_B is Boltzman's constant. Three isotopic variations of each of the two reactions are also calculated, as follows. The magnitude of the potential V_2 , which includes ZPVE corrections, is adjusted, if necessary, so that the absolute magnitude of the calculated rate matches the experimental observations for the parent perhydro systems. Then the ZPVE corrections are replaced by the ap-

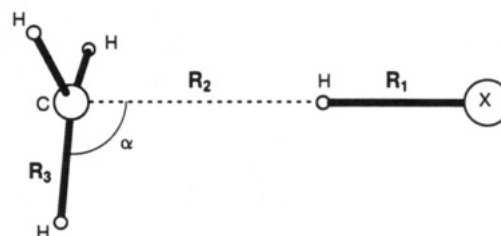


Figure 2. Representative geometry of the bound complex and transition-state structure for reactions $\text{CH}_3 + \text{HCl}$ and $\text{CH}_3 + \text{HBr}$. Refer to Table I for details.

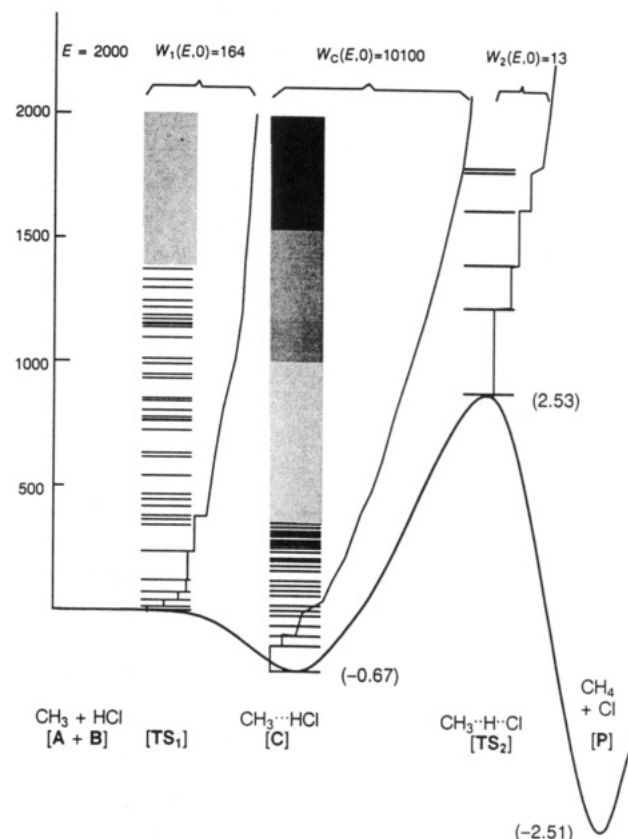


Figure 3. G1 potential energy surface for $\text{CH}_3 + \text{HCl} \rightarrow \text{CH}_4 + \text{Cl}$. The vertical energy scale is in units of reciprocal centimeters. The numbers in parentheses are the energies of stationary points in kilocalories per mole. The functions $W_1(E, 0)$, $W_c(E, 0)$, and $W_2(E, 0)$ are plotted on a logarithmic horizontal scale, and the values at $E = 2000 \text{ cm}^{-1}$ are given for illustrative purposes.

propriate calculated values for the particular isotopic substitution pattern, to derive the potential V_2 for the reaction with isotopic substitution. As it happens, no adjustment was necessary for the reactions involving $\text{X} = \text{Cl}$, since the theoretical quantities κk_{RRKM} yielded a quantitative fit to the experimental values. In the case of $\text{X} = \text{Br}$, the theoretical value of V_2 was empirically lowered by 0.78 kcal/mol as discussed below.

Results and Discussion

The structures and vibrational frequencies of the reactants, CH_3 and HCl or HBr , intermediate complexes, transition structures, and products were determined at the second-order Moller-Plesset level of theory using the 6-31G* basis set as reported in the previous paper.²² The results are listed in Tables I (geometries, see also Figure 2) and II (vibrational frequencies). The potential energy surface was determined at the GAUSSIAN-1 (G1)²⁷ level of theory.²² Schematic representations of the computed potential energy surfaces for $\text{X} = \text{Cl}$ and $\text{X} = \text{Br}$ are shown in Figures 3 and 5, respectively.

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TABLE I: Geometrical Parameters at the MP2/6-31G* Level of Theory

	structure ^a				rotational constant ^b		
	R ₁	R ₂	R ₃	α	A	B	C
CH ₃ (D _{3h})			1.078	90.0	287.457	287.457	143.730
CD ₃					143.842	143.842	71.921
CH ₄ (T _d)		1.090	1.090	109.471	158.361	158.361	158.361
CD ₄					79.242	79.242	79.242
HCl	1.280					314.885	
DCI						161.971	
CH ₃ ·HCl complex (C _{3v})	1.287	2.337	1.080	94.3	144.158	3.583	3.583
CD ₃ ·HCl complex					72.135	3.117	3.117
CH ₃ ·DCI complex					144.158	3.582	3.582
CD ₃ ·DCI complex					72.135	3.117	3.117
CH ₃ HCl TS ₂ (C _{3v})	1.431	1.438	1.084	101.2	147.720	5.545	5.545
CD ₃ HCl TS ₂					73.917	4.743	4.743
CH ₃ DCI TS ₂					147.720	5.527	5.527
CD ₃ DCI TS ₂					73.917	4.735	4.735
HBr	1.435					246.70	
DBr						124.996	
CH ₃ ·HBr complex (C _{3v})	1.445	2.264	1.080	94.5	144.244	2.845	2.845
CD ₃ ·HBr complex					72.177	2.417	2.417
CH ₃ ·DBr complex					144.244	2.834	2.834
CD ₃ ·DBr complex					72.177	2.410	2.410
CH ₃ HBr TS ₂ (C _{3v})	1.521	1.618	1.083	99.0	146.159	3.890	3.890
CD ₃ HBr TS ₂					73.134	3.273	3.273
CH ₃ DBr TS ₂					146.159	3.860	3.860
CD ₃ DBr TS ₂					73.136	3.255	3.255

^a Refer to Figure 2 for definition of parameters: bond lengths in angstroms and bond angles in degrees. ^b In units of gigahertz.

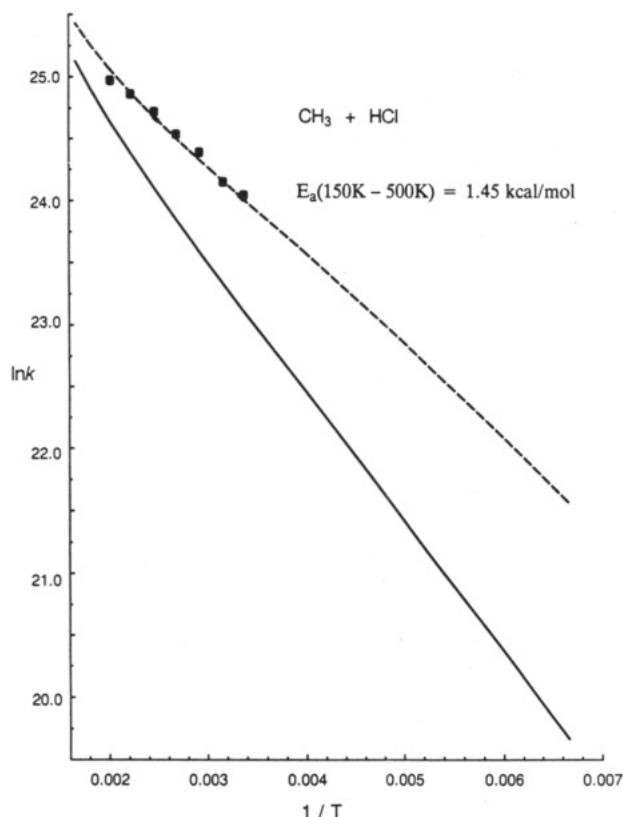


Figure 4. Arrhenius plots for CH₃ + HCl → CH₄ + Cl (V₂ = 2.53 kcal/mol): k_{RRKM}, —; κk_{RRKM}, ---; expt,²⁸ ■.

Rate Constant and Activation Energy. The RRKM rate constant k_{RRKM} is obtained by direct numerical integration of eq 22 for the calculated barrier height V₂ and several values above and below it, over the temperature range 150–600 K, for both the reactions CH₃ + HCl and CH₃ + HBr and several deuterated analogues. The results are listed in Tables III–VI and displayed as Arrhenius plots (log k vs 1/T) in Figures 4 and 6, respectively. It is immediately evident that the CH₃ + HBr reaction has strongly curved Arrhenius plots and displays a “negative” activation barrier in all cases when the barrier is low. The two halo systems are discussed separately below.

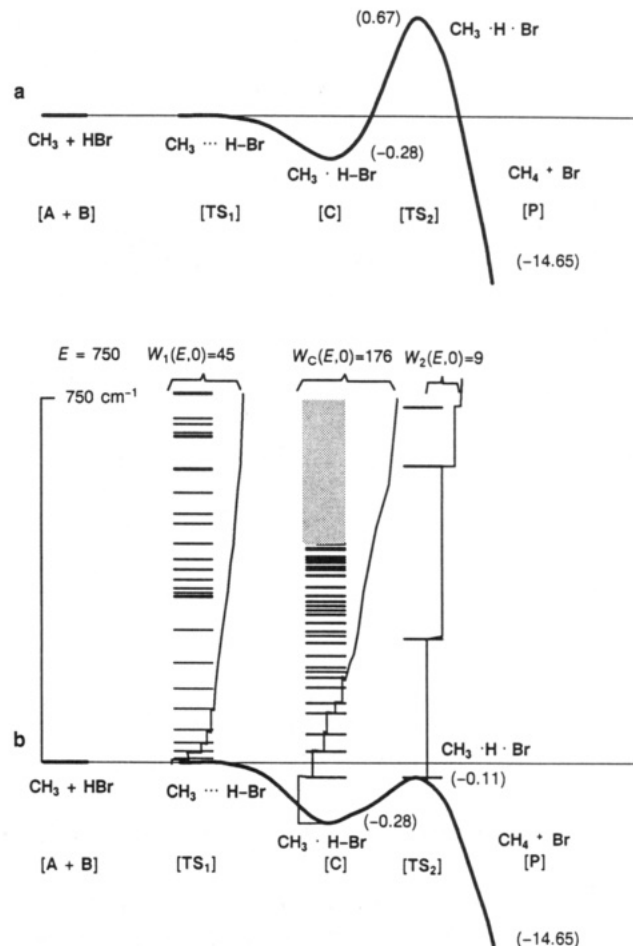


Figure 5. Potential energy surface for CH₃ + HBr → CH₄ + Br: (a) G1 level; (b) fitted to experiment. The vertical energy scale is in units of reciprocal centimeters. The numbers in parentheses are the energies of stationary points in kilocalories per mole. The functions W₁(E,0), W_c(E,0), and W₂(E,0) are plotted on a logarithmic horizontal scale, and the values at E = 750 cm⁻¹ are given for illustrative purposes.

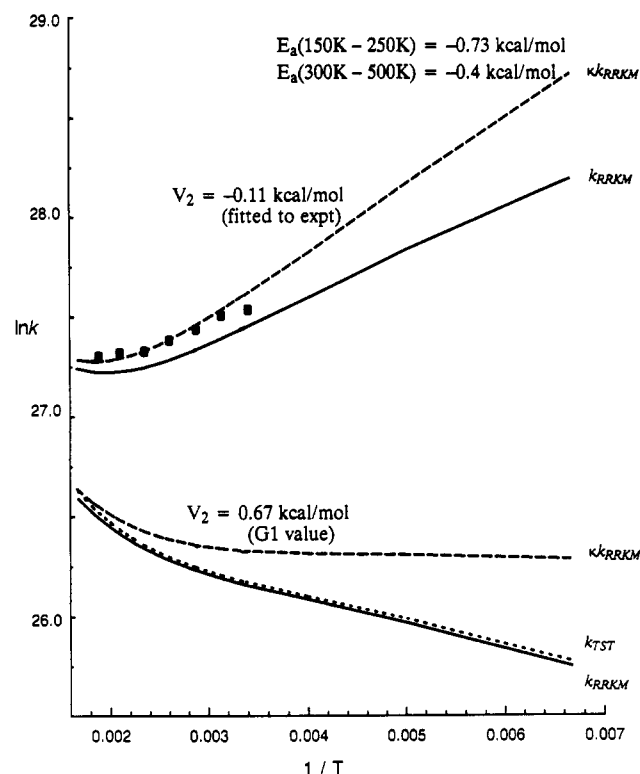
CH₃ + HCl. Schematic representations of the potential energy surface along the reaction coordinate are shown in Figure 3. Only the relative energies of the stationary points are meaningful; the

TABLE II: Vibrational Frequencies^a at the MP2/6-31G* Level

symm	CH ₃	CD ₃	symm	CH ₄	CD ₄	HCl	DCI	HBr	DBr
A ₁ '	3059	2164	A ₁	2956	2091	2896	2077	2540	1808
A ₂ ''	386	299	E	1544	1092				
E'	3239	2414	T ₂	3087	2287				
E'	1407	1035	T ₂	1343	1015				

Complexes								
symm	CH ₃ HCl	CD ₃ HCl	CH ₃ DCI	CD ₃ DCI	CH ₃ HBr	CD ₃ HBr	CH ₃ DBr	CD ₃ DBr
A ₁	3049	2784	3049	2158	3049	2396	3049	2158
	2184	2157	1997	1997	2396	2158	1706	1706
	599	463	599	463	618	476	618	476
	95	90	95	89	86	80	86	80
E	3227	2405	3227	2405	3227	2405	3227	2405
	1407	1034	1407	1034	1406	1034	1406	1034
	312	300	242	222	330	315	261	235
	135	100	124	96	148	111	134	106

Transition Structures (TS ₂)								
symm	CH ₃ HCl	CD ₃ HCl	CH ₃ DCI	CD ₃ DCI	CH ₃ HBr	CD ₃ HBr	CH ₃ DBr	CD ₃ DBr
A ₁	3015	2142	3015	2141	3027	2146	3027	2146
	1165	968	1113	910	1299	1256	1072	984
	520	458	508	453	763	570	700	551
	1218i	1208i	933i	928i	427i	424i	402i	398i
E	3178	2365	3178	2365	3198	2381	3198	2381
	1409	1033	1409	1033	1406	1032	1406	1032
	904	838	748	652	737	689	606	529
	366	285	317	262	327	251	284	233

^a Frequencies (cm⁻¹) are calculated at the MP2/6-31G*//6-31G* level and scaled by 0.95.Figure 6. Arrhenius plots for CH₃ + HBr → CH₄ + Br: k_{TST} , —; k_{RRKM} , ---; κk_{RRKM} , ···; expt,¹⁰ ○.

horizontal axis is arbitrary. The transition-state rate constant k_{TST} (eq 24) and the RRKM rate constant k_{RRKM} , which is obtained by direct numerical integration of eq 25, were evaluated at the G1 barrier height, $V_2 = 2.53$ kcal/mol, and several values above and below it, over the temperature range 150–600 K. The results for the G1 value combined with the Wigner tunneling correction²⁶ and the experimentally determined values²⁸ are listed in Table III and displayed as Arrhenius plots ($\ln k$ vs $1/T$) in Figure 4.

TABLE III: Rate Constants^a for the Reaction CH₃ + HCl → CH₄ + Cl ($V_2 = 2.53$ kcal mol⁻¹)

T, K	k_{TST}	κk_{TST}^b	k_{RRKM}	κk_{RRKM}^b	expt ^c
150	0.00348	0.0232	0.00338	0.0226	
200	0.0202	0.0848	0.0198	0.0831	
250	0.0569	0.173	0.0559	0.170	
296	0.108	0.267	0.107	0.262	0.28
300	0.114	0.275	0.112	0.271	
316	0.136	0.309	0.134	0.305	0.31
342	0.175	0.367	0.173	0.362	0.39
350	0.188	0.385	0.186	0.379	
372	0.227	0.437	0.223	0.430	0.45
400	0.280	0.504	0.276	0.496	
406	0.292	0.519	0.288	0.511	0.54
450	0.388	0.634	0.381	0.622	0.63
495	0.500	0.761	0.489	0.744	0.70
500	0.513	0.776	0.502	0.758	
550	0.655	0.932	0.637	0.907	
600	0.816	1.106	0.788	1.068	

^a Units: cm³ mol⁻¹ s⁻¹ × 10¹¹. ^b Rate including tunneling factor κ . ^c Reference 28.TABLE IV: Rate Constants^a for Reactions Involving Deuteration

T, K	CD ₃ + HCl, $V_2 = 1.91^b$		CH ₃ + DCI, $V_2 = 3.02^b$		CD ₃ + DCI, $V_2 = 2.39^b$	
	k_{TST}	κk_{TST}^c	k_{TST}	κk_{TST}^c	k_{TST}	κk_{TST}^c
150	0.0139	0.0915	0.000356	0.00155	0.00149	0.0066
200	0.0500	0.207	0.00324	0.00931	0.00824	0.0242
250	0.107	0.323	0.0121	0.0266	0.0232	0.0519
296	0.174	0.423	0.0278	0.0517	0.0452	0.0852
300	0.180	0.432	0.0296	0.0543	0.0475	0.0884
316	0.206	0.467	0.0373	0.0653	0.0574	0.102
342	0.252	0.523	0.0521	0.0855	0.0756	0.126
350	0.267	0.541	0.0572	0.0923	0.0818	0.133
372	0.309	0.590	0.0729	0.112	0.100	0.156
400	0.366	0.654	0.0960	0.141	0.126	0.188
406	0.379	0.668	0.101	0.148	0.133	0.195
450	0.479	0.777	0.147	0.201	0.182	0.252
495	0.593	0.898	0.204	0.266	0.243	0.320
500	0.607	0.912	0.210	0.274	0.250	0.328
550	0.750	1.059	0.288	0.359	0.332	0.417
600	0.911	1.229	0.380	0.460	0.427	0.519

^a Units: cm³ mol⁻¹ s⁻¹ × 10¹¹. ^b Units: kcal mol⁻¹. ^c Rate including tunneling factor κ .

TABLE V: Calculated Rate Constants^a for CH₃ + HBr → CH₄ + Br

T, K	$V_2 = 0.67 \text{ kcal mol}^{-1}$				$V_2 = -0.11 \text{ kcal mol}^{-1}$ ^b			exptl ^c
	k_{TST}	κk_{TST}^d	k_{RRKM}	κk_{RRKM}^d	k_{TST}	k_{RRKM}	κk_{RRKM}^d	
150	1.58	2.68	1.54	2.61	21.59	17.51	29.74	
200	1.95	2.72	1.91	2.67	13.90	12.23	17.03	
250	2.18	2.73	2.14	2.68	10.48	9.63	12.05	
296	2.34	2.76	2.31	2.72	8.82	8.30	9.78	9.2 ^e
300	2.36	2.77	2.32	2.73	8.72	8.21	9.65	
319	2.42	2.79	2.38	2.75	8.28	7.85	9.06	8.9
348	2.52	2.84	2.48	2.80	7.78	7.43	8.39	8.3
350	2.52	2.85	2.49	2.80	7.75	7.40	8.35	
385	2.65	2.93	2.61	2.88	7.35	7.06	7.81	7.8
400	2.71	2.97	2.66	2.92	7.22	6.95	7.64	
427	2.81	3.06	2.76	3.00	7.05	6.81	7.39	7.4
450	2.91	3.14	2.85	3.08	6.96	6.72	7.25	
478	3.03	3.24	2.97	3.17	6.90	6.66	7.12	7.3
500	3.14	3.34	3.07	3.26	6.88	6.64	7.06	
532	3.30	3.48	3.21	3.39	6.90	6.65	7.02	7.2
550	3.40	3.57	3.30	3.47	6.93	6.67	7.02	
600	3.68	3.84	3.55	3.71	7.08	6.78	7.07	

^a Units: cm³ mol⁻¹ s⁻¹ × 10¹¹. ^b Fitted to experiment.¹⁰ ^c Reference 10. ^d Rate including tunneling factor κ . ^e Average value.

TABLE VI: Rate Constants^a for Reactions Involving Deuteration

T, K	CD ₃ + HBr, $V_2 = -0.66^b$		CH ₃ + DBr, $V_2 = 0.02^b$		CD ₃ + DBr, $V_2 = -0.52^b$	
	k_{RRKM}	κk_{RRKM} ^c	k_{RRKM}	κk_{RRKM} ^c	k_{RRKM}	κk_{RRKM} ^c
150	21.23	35.85	7.18	11.63	10.25	16.47
200	14.62	20.28	5.46	7.37	7.33	9.84
250	11.28	14.08	4.64	5.68	5.93	7.22
296	9.53	11.22 (10.9) ^d	4.26	4.94	5.23	6.04
300	9.42	11.04	4.24	4.90	5.19	5.97
319	8.93	10.29	4.16	4.72	5.01	5.68
348	8.34	9.41	4.08	4.54	4.81	5.35
350	8.31	9.36	4.07	4.53	4.80	5.33
385	7.80	8.62	4.04	4.42	4.65	5.08
400	7.63	8.37	4.04	4.40	4.61	5.01
427	7.39	8.02	4.07	4.38	4.57	4.91
450	7.23	7.78	4.11	4.40	4.56	4.87
478	7.08	7.56	4.18	4.44	4.57	4.84
500	6.99	7.42	4.25	4.49	4.59	4.84
532	6.90	7.28	4.36	4.58	4.64	4.87
550	6.86	7.22	4.44	4.64	4.68	4.89
600	6.82	7.11	4.67	4.85	4.81	4.99

^a Units: cm³ mol⁻¹ s⁻¹ × 10¹¹. ^b Units: kcal mol⁻¹. ^c Rate including tunneling factor κ . ^d Experiment.¹⁰

It is apparent that for the CH₃ + HCl reaction the calculated rate constants including the Wigner tunneling correction are in excellent agreement with the experimental values²⁸ over the entire temperature range. The theoretical model also predicts that in the lower temperature region (200–500 K) the Arrhenius plot shows nearly a straight line, with slope corresponding to $E_a = 1.45$ kcal/mol (experimental, 1.4 ± 0.3 kcal/mol²⁸). However, the line starts to curve upward above 500 K.

The “active” energy levels (rotation and vibration) of TS₁, and vibrational energy levels of the intermediate complex and of TS₂, are shown in Figure 3. The functions $W_1(E,0)$ and $W_2(E,0)$ are plotted on a logarithmic scale to show their relative magnitudes. The transition-state theory rate constants is within a few percent of the RRKM rate constant over the entire temperature range, a consequence of the small branching ratio (eq 9) at all values of energy above the barrier. The corresponding function for the complex $W_C(E,0)$ is also shown. The much larger number of “active” states of the complex than of either TS₁ or TS₂ supports the current model in which it is postulated that once formed, the complex “forgets” its origins by internal energy reorganization.

Isotope Effects. For the reactions involving deuterium substitution, the rate constants were evaluated as κk_{TST} . The value of V_2 for each deuterated reaction is calculated by assuming that it has the same G1 Born–Oppenheimer potential surface as the undeuterated system and then adding the appropriate zero-point vibrational energy. The results are listed in Table IV. No experimentally determined isotope data are available. Each of the

effective values of V_2 are similar in magnitude to the parent system. At 296 K, the isotopic effects are $k(\text{CH}_3+\text{HCl})/k(\text{CH}_3+\text{DCI}) = 5.08$ (primary isotopic effect), $k(\text{CH}_3+\text{HCl})/k(\text{CD}_3+\text{HCl}) = 0.62$ (secondary isotopic effect), and $k(\text{CH}_3+\text{HCl})/k(\text{CD}_3+\text{DCI}) = 3.08$ (mixed primary and secondary effects). Thus, the reaction with HCl is predicted to exhibit a normal primary isotope effect and an inverse secondary isotope effect.

CH₃ + HBr. Schematic representation of the potential energy surface along the reaction coordinate as calculated by G1 theory is shown in Figure 5a. The transition-state rate constant k_{TST} (eq 24) and the RRKM rate constant k_{RRKM} (eq 25) were evaluated at the G1 barrier height, $V_2 = 0.67$ kcal/mol, combined with the Wigner tunneling correction,²⁶ and plotted as a function of temperature in Figure 6 (lower curves), together with the experimental data.¹⁰ The data are also listed in Table V. The theoretical values of κk_{RRKM} are approximately 1 order of magnitude lower than the experimentally measured rate constants and exhibit a positive temperature dependence, contrary to the experimental observations. The A factor of the Arrhenius equation can be evaluated by

$$A = \kappa \frac{e^2 k_B T}{h} \exp(\Delta S_C^* / R) \quad (24)$$

where ΔS_C^* is the entropy difference between TS₂ and reactants at standard concentration (1.0 mol/L), k_B is Boltzman's constant, and κ is the tunneling correction. The value of ΔS_C^* is calculated from the data in Tables I and II. The A factor so derived is 9.5×10^{-13} cm³ molecule⁻¹ s⁻¹, in close agreement with the experimental value, 8.7×10^{-13} cm³ molecule⁻¹ s⁻¹.¹⁰ Therefore, it is likely that the deviation of the theoretical rate constants from the experimental results is mainly due to the overestimation of V_2 . Consequently, the rate constants were recalculated at lower values of V_2 , assuming that the value of the imaginary frequency does not change. The best fit to the experimentally measured data was achieved for $V_2 = -0.11$ kcal/mol, an adjustment of less than 1 kcal/mol from the theoretical value. The resulting potential curve is shown in Figure 5b, and the Arrhenius plots are displayed in Figure 6 (upper curves). Of special interest is the fact that the predicted curvature of the Arrhenius plot matches well the experimental data over the whole temperature range. The theoretical model predicts that, for any barrier V_2 below +0.67 kcal/mol, negative temperature dependence of κk_{RRKM} and hence an apparent negative activation energy would be observed, at least in the lower temperature region (150–250 K). In light of the theoretical values, it is apparent that fitting a straight line to the theoretical curve or to the experimental data points in the temperature range 300–500 K yields a rather arbitrary value for the Arrhenius activation energy (Figure 6). It should also be noted that, according to the model, the temperature dependence becomes positive at sufficiently high temperatures even when the depen-

dence at lower temperatures is negative.

As discussed above in connection with Figure 3, the "active" energy levels of TS_1 , the intermediate complex, and TS_2 are shown in Figure 5b. The functions $W_1(E,0)$ and $W_2(E,0)$ and the corresponding function for the complex $W_C(E,0)$ are also shown. It is apparent that, even with the lower forward barrier, there are many more available channels for the reverse reaction so that the branching ratio remains relatively small. Nevertheless, there is a more significant difference between k_{TST} and k_{RRKM} (see also Table V) than is encountered with larger values of V_2 .

Isotope Effects. For the reactions involving deuterium substitution, the rate constants were evaluated as k_{RRKM} . The value of V_2 for each deuterated reaction is calculated by assuming that it has the same fitted Born-Oppenheimer potential surface as the undeuterated system. The results are listed in Table VI. Each of the effective values of V_2 are close to zero, ranging from -0.66 to +0.02 kcal/mol. Negative temperature dependence is predicted in each case over most of the temperature range investigated. At 296 K, the isotopic effects are $k(CH_3+HBr)/k(CH_3+DBr) = 1.98$ (primary isotopic effect), $k(CH_3+HBr)/k(CD_3+HBr) = 0.87$ (secondary isotopic effect), and $k(CH_3+HBr)/k(CD_3+DBr) = 1.62$ (mixed primary and secondary effects). The calculated inverse secondary isotope effect, 0.87, is in excellent agreement with the experimentally determined value, 0.83 ± 0.08 .¹⁰ The reaction with HBr, as with HCl, is predicted to exhibit a normal primary isotope effect ($k_H/k_D > 1$), albeit smaller than in the case of HCl. McEwen and Golden have suggested that the magnitude of the primary isotope effect may be used to distinguish between the two possible mechanisms for free-radical hydrogen abstraction reactions. They postulated that a mechanism involving an intermediate complex (the chemical activation mechanism) should exhibit an inverse kinetic isotope effect ($k_H/k_D < 1$) as found in their modeling of the reaction of $t-C_4H_9$ with HI. A simple metathesis reaction is expected to exhibit a normal isotope effect.

The present results, which clearly are based on the chemical activation model and encompass both positive ($X = Cl$) and negative ($X = Br$) temperature dependence, indicate that the distinction between reaction mechanisms cannot be made on the basis of isotope effects. A normal primary isotope effect is predicted in either case.

Conclusions

Both reactions $CH_3 + HX \rightarrow CH_4 + X$ ($X = Cl, Br$) proceed via a loosely bonded complex which is formed without activation energy according to G1 theory. Rate constants for the reaction with $X = Cl$, calculated based on two-channel RRKM theory with corrections for tunneling evaluated using the Wigner method, agree quantitatively with the experimental data. In the case of the system with $X = Br$, a match to the experimental data could only be found if the G1 value of the barrier is lowered by 0.78 kcal/mol to -0.11 kcal/mol. The discrepancy between the G1 value for the barrier and that required to match experiment is well within the range of errors expected for G1 theory. A normal primary isotope effect ($k_H/k_D > 1$) is predicted for each reaction, while secondary isotope effects are predicted to be inverted ($k_H/k_D < 1$). Specifically, the values at 296 K for k_H/k_D are as follows: $CH_3 + DX$, 5.08 ($X = Cl$), 1.98 ($X = Br$); $CD_3 + HX$, 0.62 ($X = Cl$), 0.87 ($X = Br$); $CD_3 + DX$, 3.08 ($X = Cl$), 1.62 ($X = Br$). Quantitative agreement with the single experimental value at 296 K for $CD_3 + HBr$ vs $CH_3 + HBr$, $k_H/k_D = 0.83 \pm 0.08$, supports the present theoretical description of the reaction mechanisms.

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A Statistical Thermodynamical Description of the Cation Distribution and Ion Exchange in Zeolites

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A statistical thermodynamical model is proposed for describing the temperature-dependent equilibrium distribution of different types of cations among the crystallographically different sublattices in zeolites. Thermodynamic functions relating to ion exchange involving several groups of homogeneous sites can be derived. On the basis of the Gibbs-Duhem relation, it is shown that no standard chemical potentials, and hence no specific ion-exchange equilibrium constants, of a cation in a site of a zeolite can be defined.

1. Introduction

A characteristic aspect of a zeolite framework is that it often provides more sites than the number of cations required to neutralize the anionic charge. The equilibrium distribution of these cations among the different sites depends on several factors¹ such as the framework topology and composition, the presence of adsorbates, the charge and size of the cations, and the temperature. Also, when comparing structures with different cation types, one

may notice that the number of cations per group of sites is also subject to change; this complicates the theoretical description of ion exchange in zeolites.

Recently, Van Dun² proposed a statistical thermodynamical model for describing the temperature-dependent distribution of one type of cation among different groups of sites. This model was verified experimentally.³ Along the same lines, a model for the multication equilibrium distribution can be developed. At the

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